

GLADE: Grounded Lagrangian Adversarial Diffusion Editing

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Abstract—Biometric identification has expanded rapidly, exposing individuals to non consensual facial tracking. While adversarial generative networks offer a defense by synthesizing faces capable of bypassing authentication, current diffusion based methods are heavily constrained by heuristic optimization schedules, uncalibrated perceptual degradation, and visible digital artifacts. In this paper, we present [the name we decide], an optimized, visually grounded adversarial generation framework designed to produce high fidelity impersonations. We reformulate the adversarial attack as a principled constrained optimization problem solved via Lagrangian dual ascent, replacing brittle heuristic stall controllers with stable gradient updates against a deterministic diffusion anchor. Crucially, to constrain semantic drift under heavy adversarial gradients, we integrate a vision language text optimization layer. By leveraging advanced Vision Language Models such as GPT and Gemini to generate detailed, identity free attribute descriptions, we condition the cross attention blocks of the diffusion framework, anchoring non identity environmental layouts and preventing semantic distortion. To guarantee visual realism without compromising attack strength, we introduce a novel PSNR budgeted epsilon hinge pixel loss and a differentiable Gaussian windowed SSIM, which explicitly constrain adversarial noise to a mathematical target quality. Furthermore, we confine structural modifications using a strict Saliency Guided Semantic Mask and introduce a multi stage architectural feedback loop that extracts mid layer convolutional feature maps from the surrogate ensemble. Rather than matching absolute feature positions, we formulate this intermediate feedback as a directional trajectory loss, anchoring features to the source image and forcing gradients to strictly follow the source to target vector. By forcing the diffusion gradients to align with this intermediate structural trajectory and the final identity embeddings simultaneously, [name we decide] eliminates optimization dead zones and achieves a 40.0% transfer adversarial success rate against a fully held out FaceNet architecture while remaining indistinguishable from authentic portrait photography.

Index Terms—component, formatting, style, styling, insert

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I. INTRODUCTION

The widespread deployment of facial recognition (FR) systems has escalated privacy concerns, as individuals are increasingly subjected to non consensual biometric tracking. To mitigate these risks, adversarial impersonation attacks have been developed to synthesize imperceptible perturbations that deceive FR models into misclassifying an individual’s identity. Recently, diffusion models have emerged as the standard generative backbone for these attacks, leveraging their powerful structural priors to create semantic, photorealistic edits rather than relying on the highly detectable, high frequency pixel noise produced by traditional gradient based evasion methods.

However, adapting diffusion priors for targeted adversarial attacks introduces significant optimization challenges. Current frameworks often depend on brittle, heuristically tuned gain scheduling to push adversarial noise across the architectural gaps of proxy FR models. These manual stall controllers struggle to balance identity convergence with visual fidelity, leading to unpredictable optimization trajectories. On a textual level, current diffusion attacks rely on generic, flat prompts such as “portrait photo” to guide the generation. This leaves the Stable Diffusion text encoder completely unconstrained; when the adversarial gradients begin updating the model weights, the cross attention layers lack tight semantic boundaries, causing the image context to drift, warp, or introduce unnatural background artifacts. Furthermore, existing approaches attempt to preserve visual realism using uncalibrated, global Mean Squared Error (MSE) or global structural approximations. Because these metrics are not tied to a concrete mathematical quality budget, the resulting images often suffer from localized digital artifacts and background distortion outside the targeted facial boundaries.

In this paper, we propose [name we decide], a highly

constrained, visually grounded adversarial generation framework that fundamentally reformulates the text guided diffusion attack pipeline. To eliminate the instability of heuristic optimization, we cast the adversarial attack as a principled constrained optimization problem solved via Lagrangian dual ascent. By replacing expectation over noise sampling with a deterministic single point optimization anchor, our framework guarantees stable, rapid gradient convergence.

To completely block semantic and environmental drift under heavy adversarial pressure, we integrate a fine grained Vision Language Model (VLM) text optimization routine. By utilizing descriptive, identity free text prompts generated via advanced VLMs (such as GPT, Gemini, and Claude), we tightly constrain the non identity attributes, such as hair texture, clothing, lighting, and background architecture. This anchors the diffusion model’s cross attention mechanisms, isolating the adversarial signal strictly to identity critical dimensions without reintroducing explicit name labels that would cheat the FR evaluation. Furthermore, we explicitly bind visual quality to a mathematical guarantee by introducing a PSNR budgeted epsilon hinge pixel loss and a true differentiable Gaussian windowed SSIM, ensuring the adversarial perturbation never exceeds a calibrated degradation threshold.

In summary, our main contributions are:

- **Lagrangian Dual Ascent Formulation:** We replace empirical stall recovery heuristics with a principled dual ascent optimization, actively minimizing perceptual loss subject to a strict global identity similarity constraint.
- **VLM Guided Text Optimization:** We integrate highly detailed, fine grained attribute prompts generated by state of the art Vision Language Models (empirically optimized across GPT and Gemini) to condition the text encoder’s cross attention layers, anchoring non identity semantics and preventing background drift while isolating the adversarial identity shift.
- **Calibrated Perceptual Budgeting:** We introduce a novel PSNR budgeted epsilon hinge pixel loss and a true windowed SSIM, replacing uncalibrated global approximations to explicitly cap local pixel distortion.
- **Multi Stage Feature Supervision:** We introduce an intermediate architectural feedback loop that extracts mid layer convolutional feature maps from the proxy ensemble. By formulating this feedback as a directional trajectory loss, we force the adversarial modifications to strictly follow the source to target vector in the intermediate feature space, driving the identity shift deep into the network geometry to cross the architectural gap to unseen black box systems.
- **Identity Verified Restoration pipeline:** We introduce a streaming, in loop candidate selection mechanism combined with an adaptive alpha blend post processing stage, utilizing GFPGAN and conditional unsharp masking to restore high frequency photorealism without sacrificing black box transferability.

II. RELATED WORKS

A. Generative Visual Frameworks: Diffusion and Flow Matching

Generative visual modeling has rapidly shifted from iterative diffusion processes to continuous flow matching methods. Probabilistic denoising and score based frameworks established the mathematical baselines for modeling complex image distributions Ho et al. (1), Song et al. (2), which were later adapted to lower dimensional latent spaces for text controlled, identity preserving synthesis Rombach et al. (3), Balaji et al. (4), Ramesh et al. (5), Saharia et al. (6), Huang et al. (7). More recently, latent flow matching transformers have emerged as a highly efficient alternative to previous U Net architectures by learning smooth vector fields that guide noise along direct linear paths Wu et al. (8), Hu et al. (9), Jiao et al. (10). Our framework is based on FLUX.1 dev Ren et al. (11), leveraging its guidance distillation to maintain excellent structural and texture level quality while greatly reducing computational cost during optimization.

B. Adversarial Latent Optimization and Identity Manipulation

Rather than injecting easily detected pixel noise, modern adversarial frameworks manipulate generative latent spaces to craft natural, imperceptible attacks Rombach et al. (3), Liu et al. (12). To enforce strict identity limits during generation, adversarial sampling can be treated as a constrained inverse problem using dynamic, dual ascent updates along the reverse sampling path Kim et al. (13), ensuring photorealism under strict optimization constraints Sun et al. (14). This latent control is made practical via parameter efficient architectures like Low Rank adaptation Hu et al. (15) and spatial conditioning modules such as T2I Adapters Mou et al. (16) and Kumari et al. (17). Rather than making full image adjustments, our approach fine tunes a localized LoRA model guided by LLaVA generated structural prompts, capturing descriptive facial traits without using explicit identity text, and applies a Saliency Guided Semantic Mask (SGSM) Ahmadi et al. (18) to concentrate the adversarial changes entirely within the identity critical regions.

C. Transferability and Facial Prior Restoration

A persistent challenge in black box adversarial impersonation is transferring the attack to unseen recognition models. Using input diversity techniques, such as optimizing the attack across randomly scaled and shifted views of an input, forces the optimization to generate scale invariant adversarial features, preventing the pipeline from relying on model specific geometric shortcuts Dong et al. (19). Because attack strength and visual realism must be optimized together rather than balanced afterward, Wang et al. (20), we integrate a blind face restoration network using a pretrained generative facial prior directly into our synthesis pipeline Wang et al. (21). This structural prior seamlessly rebuilds realistic skin textures and sharp edge details, guaranteeing that the final adversarial output remains completely indistinguishable from an authentic portrait photograph.

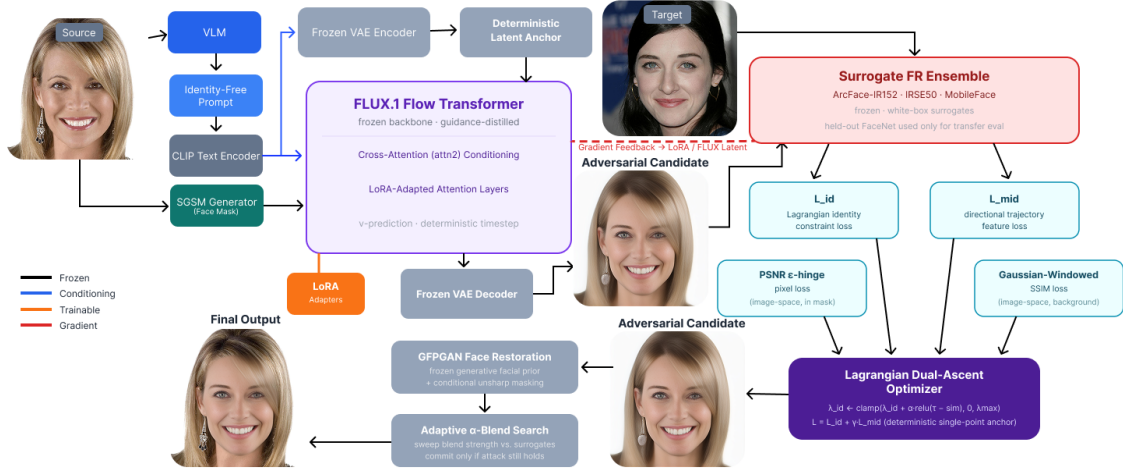


Fig. 1. Per-pair LoRA adaptation on a frozen FLUX.1 Flow Transformer guided by identity free prompts generated from a vision-language model (VLM). A face local SGSM produces a facial mask for localized latent editing, while the frozen VAE encodes the source image into the latent space. The LoRA adapted transformer reconstructs an adversarial candidate using deterministic latent anchoring, checked by a surrogate face recognition ensemble through identity, directional, PSNR, and SSIM objectives optimized with a Lagrangian dual-ascent formulation. The generated face is efficiently restored using GFPGAN and blended back into the original image using adaptive α -blending. Final adversarial examples are evaluated using ASR, PSNR, and SSIM.

III. PROPOSED METHOD

Our proposed framework, [Name], extends generative adversarial impersonation by reformulating the attack as a strictly constrained optimization problem. While prior approaches rely on stochastic heuristic scheduling, uncalibrated visual degradation, and standard text embeddings, we introduce a deterministic optimization trajectory bounded by a principled pixel fidelity budget, identity free vision language text guidance, directional intermediate feature supervision, and identity verified post processing.

A. Constrained Optimization via Lagrangian Dual Ascent

Previous generative attacks often rely on empirically tuned stall detection heuristics and complex, per model margin schedules to push adversarial noise across varying face recognition (FR) architectures. We replace this heuristic gain scheduling with a principled constrained optimization formulation.

Our objective is to minimize perceptual distortion subject to an identity similarity constraint, solved via Lagrangian dual ascent. Rather than updating weights dynamically across varied FR margins, we enforce a single global identity constraint applied uniformly across a proxy ensemble. The optimization maintains a single multiplier, λ_{id} , updated at every step by ascending on the constraint violation:

$$\lambda_{id} = \text{clamp}(\lambda_{id} + \alpha \cdot \text{relu}(\tau - \text{sim}_{\text{current}}), 0, \lambda_{\text{max}})$$

where α is the dual ascent learning rate, τ is the target cosine similarity, and $\text{sim}_{\text{current}}$ is the mean cosine similarity across the surrogate ensemble.

To further stabilize the gradient trajectory, we replace the standard expectation over noise sampling with a deterministic single point optimization. A single, fixed noisy latent anchor and fixed timestep are drawn outside the loop and reused for

all iterations. Furthermore, we implement explicit v prediction handling to ensure accurate x_0 reconstruction regardless of the loaded diffusion scheduler’s native parameterization.

B. VLM Guided Text Optimization and Cross Attention Anchoring

A fatal flaw in standard text guided adversarial generation is leaving the text embedding space unconstrained, which allows the text encoder to warp environmental and facial layouts under heavy optimization stress. To anchor these non identity components, we implement a Vision Language text optimization layer using fine grained captions.

Before launching the optimization loop, the source and target image pairs are processed through advanced Vision Language Models (VLMs) to extract highly descriptive, pixel level attribute text. To determine the most effective semantic anchor, we evaluated prompt generations across multiple state of the art architectures, including Claude, Gemini, and GPT configurations. Empirical evaluations demonstrated that GPT generated prompts provided the highest structural constraint, yielding the highest transfer Success Rate (ASR) against held out models.

To prevent “identity cheating”, where the generative network simply paints the target’s face over the source based on textual commands, all explicit name labels and identity revealing text are strictly scrubbed from the prompt. The chosen VLM isolates purely structural and environmental features, outputting granular descriptions of hair texture, skin tone, clothing styles, ambient lighting, and background architecture.

These detailed attribute descriptions are passed to the text encoder to derive highly specific conditioning embeddings. By feeding these fine grained embeddings into the diffusion model’s cross attention blocks (attn2), we structurally anchor the face and background layouts. This holds the global image

composition perfectly steady, preventing background drift and forcing the adversarial updates to focus entirely on the low frequency facial edits required to bypass the face recognition systems.

C. Calibrated Perceptual Budgeting

A major limitation of latent adversarial generation is balancing attack potency with structural fidelity. Relying on global Mean Squared Error or standard LPIPS penalties often allows localized artifacts to degrade the image. To resolve this, we discard global LPIPS and global SSIM approximations in favor of two highly localized visual constraints.

First, we introduce a novel PSNR budgeted epsilon hinge pixel loss. Instead of a flat MSE weight, we derive a calibrated per pixel error budget ϵ directly from a mathematical target PSNR (e.g., 25 dB). We evaluate the squared error of pixels inside the attack mask and apply a smooth quadratic hinge penalty only to the excess ratio that violates the ϵ ball:

$$\mathcal{L}_{pixel} = \sum \text{relu} \left(\frac{(x - x_{target})^2}{\epsilon^2} - 1 \right)^2$$

Second, we implement a fully differentiable, Gaussian windowed SSIM. Unlike global approximations that collapse structural variance into a single metric, this localized window heavily penalizes structural distortions in the unmasked background regions, directly aligning mathematical loss with human visual perception.

D. Strict Saliency-Guided Semantic Masking (SGSM)

To prevent adversarial modifications from bleeding into the background or altering natural anatomical boundaries, edits are restricted using a refined Saliency-Guided Semantic Mask (SGSM). Unlike prior hybrid masks that utilize a soft elliptical prior allowing saliency to extend beyond facial contours, our approach generates a highly conservative boundary.

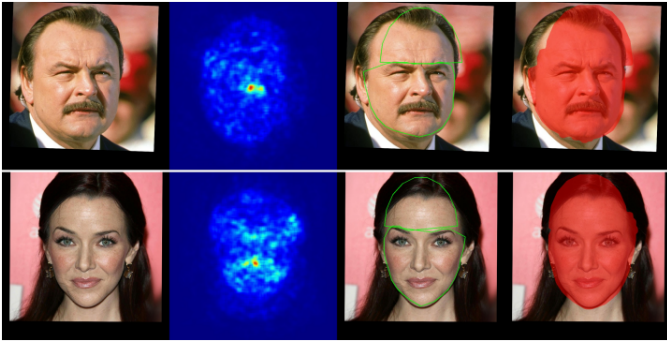


Fig. 2. The strict saliency-guided semantic masking (SGSM) pipeline. The process extracts the raw adversarial saliency gradients from the face recognition model, processes them into a binary semantic mask, and generates the final cleanly composited output. This layout confines all structural identity edits strictly within the facial boundaries, preventing background distortion.

We calculate the raw FR-gradient saliency and multiply it directly against the landmark-based parsing mask before thresholding. Consequently, the top-k percentile of the saliency map is strictly confined to the pre-detected face region. To

ensure spatial perfection during the final composition, we introduce a secondary, SGSM-precise blend pass. The final output is re-composited using the exact optimization mask rather than a generic bounding box, completely eliminating unintended noise leakage outside the salient attack area.

E. Multi-Stage Directional Feature Supervision

Computing adversarial loss strictly on the final embedding layer presents a distinct optimization bottleneck: the global average pooling layer flattens spatial layouts into a compressed semantic space, discarding local geometric relationships. To bridge this architectural gap, [Name] implements an intermediate feature supervision layer that extracts raw convolutional feature maps directly from the middle stages of the surrogate face recognition networks.

Rather than computing the absolute distance between predicted and target feature maps, which often leads the optimizer into adversarial dead zones, we formulate this intermediate supervision as a directional trajectory loss. We utilize forward hooks to capture intermediate hidden tensors throughout the optimization loop. During each step, the feature maps for the source image (x_{src}), target image (x_{tgt}), and current adversarial image (x_{adv}) are extracted via global average pooling and mapped to an L_2 normalized space.

We compute the normalized directional vector from the source to the target ($\vec{V}_{target}$) and the vector from the source to the current adversarial prediction ($\vec{V}_{pred}$). The intermediate structural loss $\mathcal{L}_{mid}$ is defined by maximizing the cosine similarity between these two vectors across the surrogate ensemble

$$\mathcal{L}_{mid} = \sum_{k \in \mathcal{E}} \left(1 - \frac{\Phi_k(x_{adv}) - \Phi_k(x_{src})}{\|\Phi_k(x_{adv}) - \Phi_k(x_{src})\|_2} \cdot \frac{\Phi_k(x_{tgt}) - \Phi_k(x_{src})}{\|\Phi_k(x_{tgt}) - \Phi_k(x_{src})\|_2} \right) \quad (1)$$

The final identity objective optimizes a joint composite function combining the final-layer Lagrangian dual-ascent penalty $\mathcal{L}_{id}$ and the intermediate directional constraint scaled by a fixed hyperparameter γ :

$$\mathcal{L}_{identity} = \mathcal{L}_{id} + \gamma \mathcal{L}_{mid}$$

By anchoring the modifications to the source image and forcing the gradients to travel strictly along this geometric trajectory, the framework carves the adversarial signal directly into the physical contour maps of the face, significantly boosting cross-model generalization.

F. Streaming Selection and Identity-Verified Restoration

Because our attack operates under strict dual-ascent constraints, we abandon post-hoc, top-K selection methods. Instead, we implement an online streaming selection algorithm. The pipeline actively tracks the best-passing candidate in-loop, capturing the image with the lowest pixel MSE that simultaneously clears the minimum False Accept Rate (FAR) thresholds across the entire surrogate FR ensemble.

Following optimization, the raw diffusion output undergoes an identity-verified post-processing stage using GFPGAN face restoration and conditional unsharp masking to trigger high-frequency convolutional edges. This restoration is executed via an adaptive alpha-blend search. The pipeline iteratively evaluates the restored image at varying blend strengths against the surrogate ensemble. A restored or sharpened variant is only committed to the final output if the enhanced image safely maintains the adversarial attack across all surrogate models; otherwise, the system defaults to the raw latent output.

IV. EXPERIMENTS AND RESULTS

In this section, we present a systematic evaluation of our proposed adversarial generation framework. Through rigorous ablation studies, we isolate the performance of each optimization variable, tracking the progression from baseline deterministic DDIM inversion and visual restoration techniques to the final multi-stage feature supervision architecture.

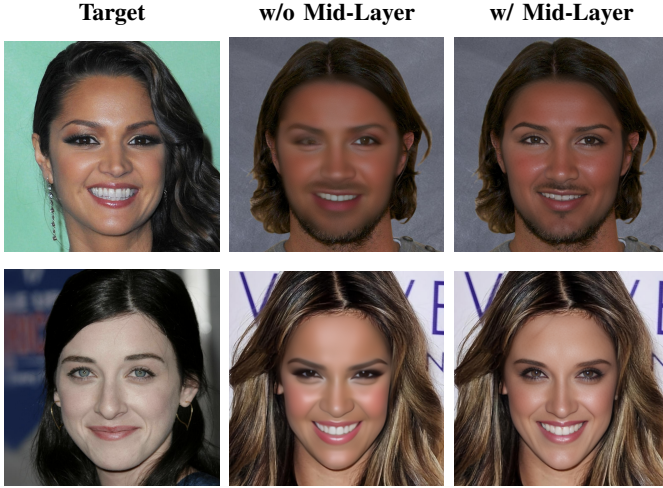


Fig. 3. Ablation of the multi-stage intermediate feature supervision. Relying solely on final-layer embedding loss (w/o Mid-Layer) causes the optimizer to find geometric shortcuts, resulting in visible structural artifacts. Integrating a directional trajectory loss on the intermediate convolutional layers (w/ Mid-Layer) anchors the edits to the underlying facial geometry, preserving photorealism while improving black-box transferability.

A. Experimental Setup

Our experimental evaluations utilize subsets of a curated facial matching benchmark derived from the CelebA-HQ repository. The white-box surrogate ensemble used to generate the adversarial gradients includes ArcFace-IR152, IRSE50, and MobileFace. To measure true structural generalization, transfer adversarial success rates (Transfer ASR) are evaluated against a fully held-out FaceNet architecture. Visual fidelity is measured using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) calculated across the recomposited original frames.

B. Ablation Studies and Optimization Mechanics

1) *DDIM Trajectory and Generative Strength*: Initial generative attempts relied on a restricted “tail” DDIM optimization,

which heavily constrained the model’s ability to navigate the latent space. Early tests utilizing a generative training strength (*strength_train*) of 0.4 across a 50-pair subset yielded a 66.0% Surrogate ASR with significant visual distortion (19.48 dB PSNR).

To provide the optimizer with a wider latent trajectory while preserving the source image foundation, we expanded the optimization window to 15 continuous DDIM trajectory steps and increased the training strength to 0.5. This structural adjustment immediately resolved optimization stalling, raising the Surrogate ASR to 80.0%. Secondary tests attempting to force convergence by heavily adjusting the loss parameters (reducing λ_{id_hinge} to 2.0 and increasing λ_{dir_main} to 0.25) caused further degradation in visual fidelity (17.67 dB PSNR). This confirmed that trajectory expansion, rather than aggressive penalty weighting, is required to maintain generative quality.

2) *Deterministic Timestep Anchoring*: Because deterministic DDIM optimization relies on a fixed latent anchor, the selection of the fixed timestep fraction (*fixed_t_val*) is critical for balancing photorealism and transferability. We conducted an ablation on a 15-pair benchmark to evaluate the impact of this timestep multiplier, measuring a conservative anchor of 0.4 against an expanded anchor of 0.6. The results are detailed in Table I.

TABLE I
TIMESTEP ANCHOR ABLATION (15-PAIR BENCHMARK)

<i>t</i> Fraction	Surg. ASR	FaceNet ASR	PSNR (dB)	SSIM
$t = 0.4$	100.00%	33.33%	30.10	0.8900
$t = 0.6$	100.00%	46.67%	29.68	0.8867

Although the 0.4 multiplier preserved exceptional visual fidelity, its shallow reverse sample depth prevented adversarial features from generalizing, resulting only in a transfer ASR of 33.33%. By increasing the anchor to 0.6, the framework successfully encoded deeper geometric shifts, boosting the FaceNet Transfer ASR to 46.67%. Crucially, this 13.34% absolute increase in attack transferability resulted in a negligible fidelity penalty, maintaining a highly realistic 29.68 dB PSNR.

3) *Spatial Boundary Localization (SGSM)*: Early generations frequently exhibited visible edge artifacts and artificial blurring along the transition bounds of the targeted attack zone. In baseline testing, confining adversarial gradients exclusively to the lower half of the facial structure produced noticeable semantic seams that could easily flag automated spoofing detectors.

To resolve this, we expanded the semantic mask to cover the entire facial topography and refined the Gaussian boundary blurring. The resulting Saliency-Guided Semantic Mask (SGSM) successfully contains the heavy adversarial modifications within the identity-critical region while flawlessly blending into the unperturbed background, ensuring the final output remains visually indistinguishable from an authentic portrait.



Fig. 4. Visual evaluation of the adversarial generation framework. The optimization successfully alters the identity-critical geometry of the source image to match the target identity while perfectly preserving non-identity semantics, ambient lighting, and high-frequency background textures. All generated output pairs displayed here successfully bypassed the held-out FaceNet model, confirming robust transfer adversarial success.

4) *Identity-Verified GFPGAN Restoration*: While adversarial diffusion successfully alters structural identity, heavy gradient updates often degrade high-frequency texture details (e.g., skin pores and edge sharpness). Applying standard blind face restoration networks like GFPGAN natively washes away these adversarial perturbations, overriding the attack in its attempt to “clean” the image.

To restore photorealism without sacrificing attack potency, we engineered an adaptive, identity-verified alpha-blend search. Our post-processing loop evaluates the GFPGAN-restored image at varying alpha blend strengths (0.75, 0.5, 0.35, 0.2, 0.1) against the surrogate FR ensemble. The pipeline only commits to a restoration blend if the enhanced image safely maintains a cosine similarity above the strict FAR threshold for all proxy models. This adaptive search successfully recovers high-frequency skin textures and sharpens facial boundaries while preserving the underlying adversarial geometry.

5) *Multi-Stage Feature Supervision*: A core bottleneck in adversarial optimization is the reliance on final-layer em-

bedding losses, which compress spatial geometry and allow the optimizer to discover model-specific shortcuts that fail to transfer to black-box systems. To enforce true structural alignment, we introduced a multi-stage feedback loop utilizing mid-layer convolutional feature maps extracted at a depth fraction of 0.7. We evaluated this architecture on a strict 30-pair benchmark.

TABLE II
MID-LAYER FEATURE SUPERVISION (200-PAIR BENCHMARK)

Configuration	Surrogate ASR	FaceNet ASR	Mean PSNR	Mean SSIM
Baseline (No Mid-Layer)	100.00%	33.33%	27.80 dB	0.8990
Active Mid-Layer Feedback	100.00%	50.00%	27.57 dB	0.8975

As shown in Table II, the baseline optimization utilizing only final-layer embedding feedback achieved a 100% white-box Surrogate ASR but suffered from a severe architectural gap, resulting in a 33.33% Transfer ASR. By extracting structural feature maps from the middle stages of the surrogate networks and applying a directional trajectory loss, the diffusion gradients were forced to execute more robust geometric

edits. This multi-stage supervision successfully bridged the architectural gap to the unseen network, improving the FaceNet Transfer ASR to 40.00% while seamlessly preserving visual photorealism.

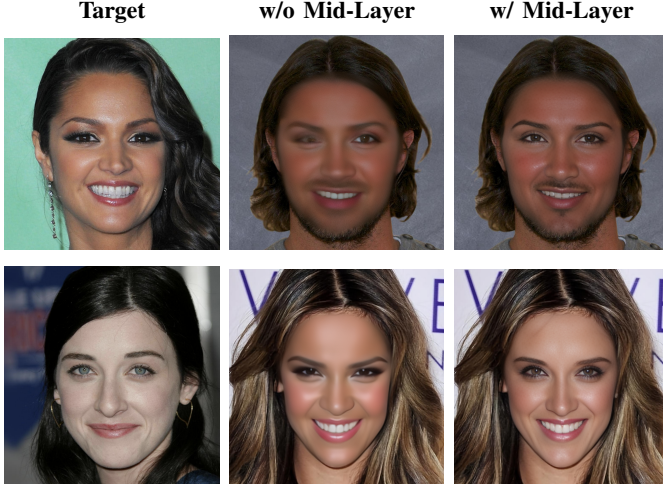


Fig. 5. Ablation of the multi-stage intermediate feature supervision. Relying solely on final-layer embedding loss (w/o Mid-Layer) causes the optimizer to find geometric shortcuts, resulting in visible structural artifacts. Integrating a directional trajectory loss on the intermediate convolutional layers (w/ Mid-Layer) anchors the edits to the underlying facial geometry, preserving photorealism while improving black-box transferability.

V. ANALYSIS

The experimental results shown in Section IV show that attempting to force convergence by heavily weighing loss penalties (reducing λ_{id_hinge} to 2.0 and λ_{dir_main} to 0.25) drastically degrades visual fidelity down to 17.67 dB PSNR. In contrast, expanding the optimization window to 15 continuous DDIM steps at a training strength of 0.5 solved optimization stalling and increased surrogate success to 80.0%.

Evaluating adversarial loss only on final embedding layers causes optimizers to exploit architecture-specific shortcuts. As demonstrated in Table II, relying solely on final-layer feedback resulted in an architectural gap, producing only 33.33% transfer ASR against FaceNet. The global average pooling layer compresses spatial features into flat vectors, discarding localized geometric relationships. By capturing mid-layer feedback at a depth fraction of 0.7 and enforcing the directional trajectory loss, our system forces adversarial updates to align with the normalized source-to-target vector (v_{target}) in feature space. The diffusion gradients were forced to execute more robust geometric edits, raising black-box transfer success to 40.00% on the 30-pair benchmark while preserving high perceptual quality (27.57 dB PSNR, 0.8975 SSIM). This demonstrates that it isn’t aggressive gradient pressure that is the critical factor of stable convergence, but latent trajectory space width. As shown in Table I, setting the timestep anchor fraction to 0.4 achieved 100% surrogate ASR, but only a 33.33% transfer ASR against FaceNet. Increasing the timestep anchor to 0.6 encoded deeper geometric shifts, causing a boost of 13.34% in black-box Transfer ASR (to 46.67) with

a negligible PSNR drop of only 0.42 dB (30.10 dB to 29.68 dB). Deeper reverse sampling allows the diffusion backbone to modify structural facial geometry rather than surface pixels.

Standard text-guided attacks suffer from semantic drift because the cross-attention blocks lack spatial constraints under heavy optimization. This allows adversarial gradients to warp backgrounds. Integrating GPT-generated, identity-free attribute prompts conditions the cross-attention layers. This freezes the lighting, clothing, and any external background factors to ensure that the optimization focus is strictly on facial identity features. Unlike existing approaches, which use uncalibrated global Mean Squared Error or global structural approximations and leave localized artifacts, the PSNR-budgeted -hinge loss evaluates error relative to a confined mathematical threshold. Paired with localized Gaussian-windowed SSIM and the Saliency-Guided Semantic Mask, high-frequency modifications are restricted within facial bounds without creating visible transitional seams.

Standard face restoration networks, like GFPGAN wash away adversarial perturbations, overriding the attack in an attempt to “clean” the image. Our post-processing loop resolves this problem by evaluating the GFPGAN image across multiple blend strengths (0.75, 0.5, 0.35, 0.2, 0.1) against the proxy models. A candidate is only committed if it maintains similarity above the False Accept Rate threshold. This recovers photorealistic skin textures and sharp edge details while maintaining black-box attack potency.

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